



King Abdul Aziz University

Faculty of Computing & Information Technology
Computer Science Department

Final Exam (August 22nd, 2017)
Course: CPCS-223 (Analysis and Design of Algorithm)

Time: 60 minutes

Student ID: _____

Student Name: _____

***Section: _____

Instructor:

☐ Dr Umair

☐

Question 11

How many linked lists are used to represent a graph with n nodes and m edges, when using an adjacency-list representation?

- a) m
- b) n
- c) $n*m$
- d) $n+m$

Question 12

What will be the order of matrix to represent a graph with m nodes and n edges, when using an adjacency-matrix representation?

- a) $n * n$
- b) $m * n$
- c) $n*m$
- d) $m * m$

Question 13

A complete graph with 5 vertices has how many edges

- a) 5
- b) 8
- c) 10
- d) None

Question 14

Multiple occurrence of the “same edge” is called:

- a) Self Edge
- b) Multi Graph
- c) Cycle
- d) None

Question 15

Maximum number of edges in a “**Directed Graph**” is calculated as:

- a) $n(n - 1) / 2$
- b) $n(n - 1)$
- c) $n(n + 1) / 2$
- d) $n(n + 1)$

Question 16

Maximum number of edges in a “**Un-Directed Graph**” is calculated as:

- a) $n(n - 1) / 2$
- b) $n(n - 1)$
- c) $n(n + 1) / 2$
- d) $n(n + 1)$

Question 17

Every “connected vertices” are also “adjacent vertices” always.

- a) True
- b) False

Question 18

In “Simple Path” all vertices will be distinct **except**:

- a) First
- b) Last
- c) First & Last both
- d) None

Question 19

How much time it will take to find out whether vertex U is adjacent to vertex V , if “**Adjacency Matrix**” is used?

- a) n
- b) 1
- c) n^2
- d) None

Question 20

How much time it will take to find out whether vertex U is adjacent to vertex V, if “**Adjacency List**” is used?

- a) n
- b) 1
- c) n^2
- d) None

Question 21

Assuming that “root” of max heap is at index “1”, how the value of parent (i) may be calculated?

- a) $(i - 1) / 2$
- b) $i - 1 / 2$
- c) $i / 2$
- d) None of above

Question 22

The condition to check **Max-Heap** property is:

- a) $A[\text{Parent}(i)] \leq A[i]$
- b) $A[\text{Parent}(i)] \neq A[i]$
- c) $A[\text{left}(i)] \leq A[i]$
- d) $A[\text{Parent}(i)] \geq A[i]$

Question 23

The condition to check **Min-Heap** property is:

- a) $A[\text{Parent}(i)] \leq A[i]$
- b) $A[\text{Parent}(i)] \neq A[i]$
- c) $A[\text{left}(i)] \leq A[i]$
- d) $A[\text{Parent}(i)] \geq A[i]$

Question 24

What is the running time of “Max-Heapify”?

- a) n
- b) $\lg n$
- c) $n \lg n$
- d) n^2

Question 25

What is the running time of “Build-Max-Heap”?

- a) n
- b) $\lg n$
- c) $n \lg n$
- d) n^2

Question 26

What is the running time of “Heap-Sort”?

- a) n
- b) $\lg n$
- c) $n \lg n$
- d) n^2

Question 27

Which of the following is not a characteristic of “Hashing”?

- a) Searching Technique
- b) No Comparison of “key” with values
- c) Running time is constant
- d) None

Question 28

Which of the following techniques use “**hash function**” to find the “**hash address**”?

- a) Direct Address Table
- b) Hash Table
- c) Open Addressing
- d) Both b & c

Question 29

Probing is a technique to address the collision problem for:

- a) Direct Address Table
- b) Hash Table
- c) Open Addressing
- d) None

Question 30

Chaining is a technique to address the collision problem for:

- a) Direct Address Table b) Hash Table
- c) Open Addressing d) None

Question 31

Problem Statement specifies the relationship between:

- a) Input & Process b) Process & Output
- c) Output & Input d) None

Question 32

Which of the following is / are not the basic characteristic of an **Algorithm**?

- a) Step by Step Method b) Unambiguous Instruction
- c) Computational Procedure d) Instances

Question 33

Which of the followings are the designing technique(s) of algorithm?

- a) Greedy Algorithm b) Decrease and Conquer
- c) Transform and Conquer d) All of above

Question 34

There are TWO phases in “problem solving”, which are:

- a) Analysis & Design b) Analysis & Synthesis
- c) Design & Synthesis d) None

Question 35

Which of the following is not the part of “Synthesis” phase in solving a problem?

- a) Building functions with supporting tools
- b) How components should be put together?
- c) How components (processes) relate to each other
- d) Composing functions to form a process

Question 36

How “**Time**” and “**Space**” are related to each other?

- a) Directly Proportional b) Inversely Proportional
- c) Both ways d) None

Question 37

Efficiency claim about an algorithm must be independent of

- a) Number of Instances b) Bench Marks
- c) Both a & b d) None

Question 38

While describing the efficiency of an algorithm, which term(s) are not ignored at all?

- a) Highest Order term b) Lower Order term
- c) All Constant d) All of Above

Question 39

Which of the following “**Complexity Class**” represents the **minimum** running time?

- a) $n \lg n$
- b) $\lg n$
- c) n
- d) n^2

Question 40

The running time of “Radix Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 41

The running time of “Selection Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 42

The running time of “Merge Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 43

The running time of “Quick Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 44

For RAM model “RAM” stands for:

- a) Random Analysis Machine
- b) Random Access Memory
- c) Random Analysis Memory
- d) Random Access Machine

Question 45

RAM model is designed primarily for

- a) Single Processor Machine
- b) Multi Processor Machine
- c) Both a & b
- d) None

Question 46

For RAM model, which one is not the basic assumption of the model?

- a) Infinitely large random-access memory
- b) Every instruction completes its execution in “unit” time
- c) Instructions do not execute sequentially
- d) Reproducible execution time

Question 47

Which analyses produce the same results most of the time?

- a) Best Case & Average Case
- b) Average Case & Worst Case
- c) Worst Case & Best Case
- d) None

Question 58

“Loop Invariant” can be defined for

- a) Iterative Algorithms
- b) Recursive Algorithm
- c) Both a & b
- d) None of above

Question 59

“Loop Invariant” can be written in

- a) Descriptive Statement
- b) Mathematical Expression
- c) Both a & b
- d) None of above

Question 60

“At the start of each iteration of the for loop the subarray $A[1 \dots j-1]$ consist of the elements originally in $A[1 \dots j-1]$ but in sorted order.” is the loop invariant for:

- a) Merge Sort
- b) Insertion Sort
- c) Quick Sort
- d) None of above

Question 61

“Recurrence” is described as

- a) Equation / Equality
- b) Inequality
- c) Both a & b
- d) None of above

Question 62

“Recurrence” is used for describing the running time of:

- a) Iterative Algorithms
- b) Recursive Algorithm
- c) Both a & b
- d) None of above

Question 63

Which of the following is / are correct for “Recurrence” $T(n) = a T(n/b) + f(n)$

- a) $T(n/b)$ define the running time of a sub-problem
- b) “a” are the number of sub-problems
- c) $f(n)$ describe the running time of all statements in algorithm except recursive calss.
- d) All of above

Question 64

Which of the following is / are incorrect for “Substitution Method”

- a) Can solve upper and lower bounds simultaneously
- b) Powerful method to solve recurrences
- c) demands practice / expertise
- d) Use Mathematical Induction to prove the guess

Question 65

Which condition is incorrect for general recurrence equation?

- a) $a \geq 1$
- b) $b > 1$
- c) $b < 1$
- d) None

Question 66

What is the value of “b” for the recurrence equation $T(n) = T(2n/3) + O(n)$

- a) $2/3$
- b) $3/2$
- c) 3
- d) $1/3$

Question 67

What is the value of “q” for $q = \lfloor (p + r)/2 \rfloor$, if p = 5 and r = 14

- a) 9
- b) 10
- c) 9.5
- d) None

Question 68

Which of the following is not a method to solve the recurrence?

- a) Mathematical Induction
- b) Substitution
- c) Master Theorem
- d) None

Question 69

Which of the following methods use “guess” to solve the recurrence?

- a) Recursion Tree
- b) Substitution
- c) Master Theorem
- d) None

Question 70

How many steps are there is “Mathematical Induction” to prove the guess?

- a) 2
- b) 3
- c) 4
- d) None

Question 71

What will be the solution of the following recurrence?

$$T(n) = T(n - 1) + n$$

- a) n
- b) lg n
- c) n²
- d) n lg n

Question 72

Which part of the “General Recurrence” $T(n) = a T(n/b) + f(n)$ will become the root of the tree?

- a) $T(n/b)$
- b) n / b
- c) $T(n)$
- d) $f(n)$

Question 73

What will be the solution of the following recurrence?

$$a) T(n) = 2T(\lfloor n/2 \rfloor + 17) + n$$

- c) n^2
- d) $n \lg n$

Question 74

The problem size of a sub-problem in the “General Recurrence” $T(n) = a T(n/b) + f(n)$ is

- a) n
- b) b
- c) n / b
- d) a

Question 75

The cost of the “recursion tree” can be determine by the sum of the cost of:

- a) all nodes
- b) all levels
- c) all sub-trees & root
- d) all of above

Question 76

In “recursion tree” method, verifying the result using “Substitution Method” is optional.

- a) TRUE
- b) FALSE

Question 77

How we can determine the “height” of a binary tree?

- a) n b) lgn
c) lgn + 1 d) None

Question 78

How we can determine the “number of levels” of a binary tree?

- a) n b) lgn
c) lgn + 1 d) None

Question 79

What will be the problem size of the recursive call for $T(n) = 2 T(n/2) + cn$ if $n = n/16$

- a) $n / 2$ b) $n / 16$
c) $n / 8$ d) $n / 32$

Question 80

What will be the “problem size” at level “i” for $T(n) = 4T(2n/3) + O(n)$?

- a)** $n / 3^i$ **b)** $n / 2^i$
c) $n / (2/3)^i$ **d)** $n / (3/2)^i$

Question 81

What will be the “cost of node” at level “i” for $T(n) = 4T(2n/3) + O(n)$?

- a)** $\text{cn} / 3^i$ **b)** $\text{cn} / 2^i$
c) $\text{cn} / (2/3)^i$ **d)** $\text{cn} / (3/2)^i$

Question 82

What will be the “number of nodes” at level “i” for $T(n) = 4T(2n/3) + O(n)$?

- a) 3^i
b) 2^i
c) 4^i
d) $(3/2)^i$

Question 83

What will be the “cost of level” “i” for $T(n) = 4T(2n/3) + O(n)$?

- a)** $\text{cn}(8/3)^i$ **b)** $4^i \text{cn}/2^i$
c) $3^i \text{cn}/(2/3)^i$ **d)** $2^i \text{cn}/(3/2)^i$

Question 84

What will be the “last level number” for $T(n) = 4T(2n/3) + O(n)$?

- a) $\log_4 n$ b) $\log_{2/3} n$
c) $\log_{3/2} n$ d) $\log_3 n$

Question 85

What will be the “cost of last level” for $T(n) = 4T(2n/3) + O(n)$?

- a) $n \log_4 3/2$ b) $n \log_4 2/3$
c) $n \log_{3/2} 4$ d) $n \log_{2/3} 4$

Question 86

What will be the value of $\sum_{i=0}^x x^i$ if $x < 1$

- a) $\frac{1}{1-x}$ b) $\frac{1}{1+x}$
c) $\frac{1}{x-1}$ d) None

Question 87

In “Matrix Chain Multiplication”, for any matrix A_i , the order of the matrix is:

- a) $p_i * p_{i-1}$ b) $p_{i-1} * p_{i-1}$
 c) $p_{i-1} * p_i$ d) $p_i * p_i$ e) None

Question 88

The recurrence $T(n) \leq 2T(n/2) + \Theta(n)$ of Quick Sort algorithm reflects:

- a) Worst Case Partitioning b) Best Case Partitioning
 c) Balanced Partitioning d) None

Question 89

Which of the following condition is incorrect for “Master Theorem Method”?

- a) $\epsilon > 0$ b) $a f(n/b) \leq c f(n)$
 c) $c < 1$ d) None

Question 90

What is the running time of Quick Sort algorithm for “Balanced Partitioning”?

- a) n b) n^2
 c) $n \lg n$ d) $\lg n$

Question 91

What is the running time of Quick Sort algorithm for “Best Case Partitioning”?

- a) n b) n^2
 c) $n \lg n$ d) $\lg n$

Question 92

Suppose $A_1 = 5 * 6$, $A_2 = 6 * 8$, and $A_3 = 8 * 9$, what will be the order of resultant matrix for optimal solution?

- a) $p_0 * p_1$ b) $p_1 * p_2$ c) $p_0 * p_3$
 d) $p_2 * p_3$ e) $p_1 * p_3$ f) None

Question 93

Suppose $A_1 = 5 * 6$, $A_2 = 6 * 8$, and $A_3 = 8 * 9$, what will be the value of p_2 ?

- a) 5 b) 6
 c) 8 d) 9 e) None

Question 94

The recurrence $T(n) = T(n-1) + \Theta(n)$ of Quick Sort algorithm reflects:

- a) Worst Case Partitioning b) Best Case Partitioning
 c) Balanced Partitioning d) None

Question 95

Suppose $A_1 = 5 * 6$, $A_2 = 10 * 5$, and $A_3 = 6 * 5$, what will be the order of resultant matrix for optimal solution?

- a) $5 * 6$ b) $10 * 5$
 c) $6 * 5$ d) $10 * 6$ e) None

Question 96

Suppose $A_1 = 5 * 6$, $A_2 = 10 * 5$, and $A_3 = 6 * 5$, which 2 matrices will be multiplied first to achieve the optimal result?

- a) A_1 & A_2 b) A_1 & A_3
 c) A_3 & A_2 d) Can't be multiplied

Question 97

Which of the following algorithm's running time depends on the maximum length of an individual instance?

- a) Quick Sort b) Bucket Sort
c) Counting Sort d) Radix Sort

Question 98

Which of the following algorithms use another sorting technique additionally to sort the data?

- a) Quick Sort b) Bucket Sort
c) Counting Sort d) Radix Sort

Question 99

Which of the following algorithms determine the frequency of each instance in given data set?

- a) Quick Sort b) Bucket Sort
c) Counting Sort d) Radix Sort

Question 100

Which of the following algorithms is primarily designed for the range $0 \leq \text{value} < 1$?

- a) Quick Sort b) Bucket Sort
c) Counting Sort d) Radix Sort

Question 101

According to the given algorithms in the book, which of the following algorithms ensure the “stability”?

- a) Quick Sort b) Bucket Sort
c) Counting Sort d) Radix Sort

Question 102

Which of the following algorithms doesn't need extra space (other than actual array) to sort the data?

- a) Quick Sort b) Bucket Sort
c) Counting Sort d) None

Question 103

Suppose $A1 = 5 * 6$, $A2 = 10 * 5$, and $A3 = 6 * 5$, what will be the optimal scalar multiplications?

- a) 600** **b) 500**
c) 400 **d) 300** **e) None**

Question 104

Which of the followings is common in both “Dynamic Programming” and “Divide and Conquer”?

- a) It stops the division when problem size is one b) Sub-problems solved separately
c) Problem is divided into sub-problems d) b & C e) All of above

Question 105

Which of the followings is the characteristic specifically associated with “Divide and Conquer”?

- a)** It stops the division when problem size is one **b)** Sub-problems solved repeatedly
c) Problem is divided into sub-problems **d)** b & C **e)** All of above

Question 106

Which of the followings is the technique which may be used to solve a problem having multiple solutions?

- a) Divide and Conquer b) Dynamic Programming
c) Brute Force Algorithm d) b & C e) All of above

Question 107

How many steps are involved in computing the “Optimal Value” using “Dynamic Programming”?

- a) 2
- b) 3
- c) 4
- d) None

Question 108

Which of the following steps is optional in “Dynamic Programming”?

- a) Characterize the structure of Optimal Solution
- b) Recursively define the value of an Optimal Solution
- c) Construct the Optimal Solution from computed information
- d) None

Question 109

Which of the followings is also known as “Optimization” technique?

- a) Divide and Conquer
- b) Dynamic Programming
- c) Brute Force Algorithm
- d) b & C
- e) All of above

Question 110

With respect to “Assembly Line Scheduling”, which option is correct?

- a) Skipping the station
- b) Vertical Transfer
- c) Re-ordering the stations
- d) Find minimum cost
- e) All of above

Question 111

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for $S_{i,j}$?

- a) Cost of station j of line i
- b) Cost till station j of line i
- c) Station j of line i
- d) Station i of line j
- e) None of above

Question 112

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for $a_{i,j}$?

- a) Cost of station j of line i
- b) Cost till station j of line i
- c) Cost of station i of line j
- d) Cost till station i of line j
- e) None of above

Question 113

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for $t_{i,j}$?

- a) Transfer cost from station j to line i
- b) Transfer cost to station j from line i
- c) Transfer cost to station j to line i
- d) Transfer cost from station j from line i
- e) None

Question 114

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for e_i ?

- a) Entry cost from line i
- b) Exit cost to line i
- c) Exit cost from line i
- d) Entry cost to line i
- e) None

Question 115

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for x_i ?

- a) Entry cost from line i
- b) Exit cost to line i
- c) Exit cost from line i
- d) Entry cost to line i
- e) None

Extra Sheet for Rough Work